Guidelines for the Use of AI in Science Fair Projects

At GARSEF we recognize the growing interest in utilizing Artificial Intelligence (AI) technology, including Large Language Models (LLMs), Machine Learning (ML) and Generative AI for research projects. However, with great power comes great responsibility. As such, we provide guidelines for responsibly engaging with AI technologies to maintain project integrity and uphold ethical standards.

Responsible Use of Generative AI Tools

- **Student Ownership:** AI tools may be used to support student learning, not replace it. All projects submitted to GARSEF must be the student's original work. ISEF added this statement about AI use to the Eligibility/Limitations section of the rulebook. "**Artificial Intelligence (AI) may be used as a resource but must be cited and given proper acknowledgment. A student is expected to do independent work and all materials presented must be in the researcher's own words. A student may not use generative AI to write the research plan, abstract, poster or to create citations (it is known to hallucinate and falsify references).**" Please note the last part in particular. Students may not use AI to write their research plan, abstract, paper, or poster. AI may only be used as a resource and must be cited. While generative AI tools are allowed, their use is optional, not mandatory.
- **Purposeful Engagement:** Students can use AI tools as aids for exploring topics, searching for information, clarifying complex texts, or checking grammar. However, students are strongly encouraged to read primary and secondary sources, analyze evidence, and make independent choices in their writing and presentations.
- **Adult Sponsor Involvement:** Students are highly encouraged to collaborate with their adult sponsors to demonstrate genuine engagement with their projects and keep them apprised of how and when they use AI

Understanding AI and LLMs

- LLMs like Chat GPT generate human-like text based on input. They operate with a knowledge cutoff (i.e. any new information, events, or developments after the tool was trained will not be accessible) and therefore may lack up-to-date information. They can provide general information but might not always be accurate or context aware.
- When using AI, avoid sharing personal details. Treat AI as a tool, not a confidant.

Citation and Verification

- Verify AI-generated information from reliable sources. You may use AI-generated information as a starting point for research.
- Ask AI for its sources and, if applicable, cite them as described below.
- Properly cite generative AI as references.

Addressing Bias

- AI may inherit biases from their training data. Critically assess AI-generated content for potential biases.
- Address identified biases in your project and work to mitigate any bias wherever possible.

Ethical Considerations

- Think about the ethical implications of using AI in your research. Uphold ethical principles in your work and discussions involving AI.
- Employ AI only for legitimate purposes. Avoid generating false information, deception, or harm.
- Use Ai tools as a supplement but rely on your independent analysis and research.
- Do not claim AI-generated essays as your own original work. Ethical concerns arise when the AI program produces essays that are claimed to be the original work of a person.
- Remember that AI is a tool to enhance your research, not a replacement for critical thinking and creativity.

AI Limitations

- LLMs such as ChatGPT may not always understand complex or context-specific questions.
- To receive accurate responses, start with specific, clear inquiries.
- Refine your questions to target the information you need and be patient and persistent in refining until you obtain accurate answers.
- Remember “garbage in = garbage out.”

Experimentation and Documentation

- Document your interactions with AI, including the questions you ask and the responses you receive. For GARSEF in 2026, we require a copy of the transcript when utilizing ChatGPT or similar AI interface. This should be added as an additional file in STEM Wizard and available at the fair for judges to review. Make sure to include the name of the AI system used and the date you utilized it.
- Maintain a logbook to ensure transparency in your research process.
- Remember, AI is a powerful tool, but it's essential to use it responsibly and ethically. Good luck with your GARSEF project!

Additional Resources

ISEF Guide - Use of generative AI to support a research project

AI vs. Machine Learning vs. Deep Learning vs. Neural Networks: What's the difference?
Video: How Large Language Models Work
Harvard Business School: Student Use Cases for AI
ChatGPT: An Introduction for Educators
Statement from the International Baccalaureate about Artificial Intelligence in Education
Ethical AI for Teaching and Learning
Shedding light on AI Bias with Real World Examples
How to Cite AI Use

AI Tools

Chat GPT
Khanmigo
Gemini
Copilot
Claude

AI Use and your Project		
Research Assistance AI may help you gather background information about your topic quickly. This may aid you in developing a strong knowledge base.	**Idea Generation** AI may assist students in brainstorming project ideas and suggesting research questions, hypotheses, or experimental design.	**Data Analysis** The use of AI may help guide students on data analysis techniques, statistical tests, or software tools for processing and interpreting results.
Experimental Procedure AI can assist in offering suggestions to improve accuracy, control variables, and ensure safety, but it can not write the research plan.	**Explanation and Clarification** Students can ask AI to explain complex scientific concepts, theories, or equations related to their projects, aiding their understanding and communication of the topic.	**Source Recommendation** AI can provide suggestions for credible sources, articles, books, or research papers. The student is expected to review the sources and create their own bibliography.
Figure Creation Students can ask for guidance on creating effective visual representations of their data, such as graphs, charts, or diagrams, to enhance the presentation of their results. All figures must be properly cited including the method used to create student images/figures.	**Ethical Considerations** AI can help students think through the ethical aspects of their projects, including considerations of human or animal subjects, environmental impact, and data privacy.	**Presentation and Communication** Students can seek tips on creating engaging project presentations, posters, or oral explanations for the science fair judges and attendees. However, students can not use AI to write the presentation for them.

The information here has been adapted from the following sources: SEFH, DMRSEF, College Board, and BRSF.

See ISEF Guidelines for AI Use to determine whether you should or should not use AI.